Evaluating Reddit Sentiment for Short-Horizon Market Prediction

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Abstract

This project examines how aggregate sentiment regarding financial assets expressed on social media communities can be used to accurately predict short-term market movements, thus generating a profitable trading strategy. We analyzed posts and comments from two Reddit subreddits. r/wallstreetbets and r/stocks, measuring sentiment using VADER (a rule and lexicon-based model). We expanded the model's dictionary with vernacular specific to the online communities for greater sentiment accuracy. Sentiment signals are then aggregated into long and short positions and evaluated over holding periods ranging from one to fourteen trading days.

We assess performance using cumulative returns, directional accuracy, and correlation with the S&P 500. While both sentiment-based strategies achieved significant cumulative returns for some holding periods, their correlation with overall market movement was low and the directional accuracy was inconsistent across time horizons. The returns were primarily driven by exposure to a small amount of volatile assets, rather than reliably predicting market movement as a whole. These results suggest that online sentiment could potentially support profitable trading strategies without providing strong signals about broad market movement.

1 Introduction

Online financial communities on social media currently play a significant role regarding how retail investors discuss markets. Platforms like Reddit have a massive amount of posts and comments where users express opinions and share trading ideas. It is often assumed that these online interactions reflect investor sentiment, thus they can seen as potential influences on market behavior. This caused interest in whether language from social media can be used to anticipate market movements. A significant challenge is separating sentiment that predicts market direction from sentiment that focuses on specific, volatile assets. Social media language is noisy and heavily depends on the community, which complicates modeling and interpretation.

This paper examines sentiment expressed in Reddit financial communities and its relationship to short-term market outcomes. We focus on two subreddits with distinct differences: r/wallstreetbets and r/stocks. Sentiment is measured using a rule and lexicon-based model designed for social media text. We seek to understand how online sentiment in social media communities predict market behavior.

2 Background

2.1 Reddit

Reddit (1) is an online community driven platform build around subreddits, or communities where users post and discuss content. Posts and comments are ranked by community votes.

2.2 r/wallstreetbets

r/wallstreetbets (2) is a subreddit community subtitled "Like 4chan found a Bloomberg Terminal." As of the time of this paper, the community consists of approximately 2.6 million self-described "degenerates" and focuses on retail stock trading and speculative investment strategies. The subreddit serves as a forum where retail investors share high-risk trading ideas, often expressed through memes, jokes, and informal language. It is noisy but is an important source of investor sentiment for the general public. This subreddit is known for its YOLO (you only live once) sentiment, referring to an aggressive, high-risk strategy where investors place typically financially unwise amounts of their portfolios into a single investment. This subreddit is most known for acts like the GameStop (GME)

short squeeze and, most recently, the Opendoor Technologies (OPEN) rally.

2.3 r/stocks

r/stocks (3) is a subreddit community subtitles "The most serious place on Reddit for Stock related discussions" consisting of 938 thousand members. Posts are focused on traditional, fundamentals-based investment discussions. It offers a structured and analytical source of investor sentiment.

2.4 VADER

VADER (4) (Valence Aware Dictionary and sEntiment Reasoner) is a lexicon and rule-based sentiment analysis specifically tuned for social media. The way VADER works is that each word in a text is looked up in VADER's sentiment dictionary. Each word in the dictionary is assigned a valence score on a scale between -4 and 4, such as great: +3.1 and terrible: -2.5. Any words not in the lexicon contribute a score of 0 to the total scores. VADER also has rules to handle modifiers, punctuation, and context. This allows VADER to account for negation ("not good" is recognized as negative valence), intensifiers ("very" after a sentiment word increases the value of the sentiment), and other rules like those. There are four scores that VADER gives: Positive (fraction of text that is positive), Negative (fraction of text that is negative), Neutral (fraction of text that is neutral), and Compound. Compound is the normalized score from -1 (most negative) to +1 (most positive) and is commonly used as the overall sentiment of the text. VADER does not require any training data, is transparent and replicable, and works with text on social media, thus making it a good choice for this project.

2.5 Simplifying assumptions

We make the following simplifying assumptions throughout our analysis:

- Language expressed in posts and comments correlates with underlying investor sentiment, rather than being purely memes or jokes.
- Sampling bias is minimal, and the collected posts and comments are representative of the communities.
- The strength of expressed sentiment in upvote engagement is proportional to the strength of the underlying intent to buy or sell the referenced asset.
- Noise introduced by spam or irrelevant content is random, approximately normally distributed, and cancels out in large samples.

3 Data

3.1 Reddit corpora

A total of 12,998 posts and comments were scraped from the following subreddits:

- **r/wallstreetbets**: "What Are Your Moves Tomorrow" threads.
- **r/stocks**: "Daily Discussion" threads.

Data were collected by appending `.json` to each Reddit thread URL to retrieve its JSON representation. From these files, we extracted post and comment text along with their timestamps, and aggregated entries into daily corpora for downstream sentiment analysis and backtesting.

3.2 Market data

Historical price data were obtained using the Python `yfinance` API. We consider two benchmark assets:

- **SPY**: a proxy for the S&P 500–an index tracking the 500 leading companies in US stock exchanges
- **MAG7**: an equal-weighted portfolio consisting of AAPL, MSFT, GOOGL, AMZN, NVDA, META, and TSLA, which are seven of largest and most influential technology companies in the US market.

3.3 NASDAQ symbol set

In this project, we use NASDAQ tickers as a reference set because most stocks discussed on Reddit are NASDAQ-listed, and this constraint reduces false positives when extracting ticker symbols from text. This dataset was obtained by downloading a complete list of NASDAQ tickers.

4 Methods

4.1 Tuning VADER

To better capture the unique language used in retail trading communities (specifically in

r/wallstreetbets), we extended the standard VADER dictionary with a custom wallstreetbets lexicon, adding prevalent slang words. This is because the basic VADER model either misinterprets or is missing sentiment for subreddit-specific vernacular. The examples below illustrate how the enhanced lexicon alters sentiment classification relative to the default VADER model.

Example 1 *Text:* "Bro $NUAI is ripping. Bought in at $0.90 and it's going parabolic in PM. Lmao."

- **Standard VADER**: Positive (compound score: $+0.599$)
- **WSB-enhanced VADER**: Positive (compound score: $+0.784$)

Example 2 *Text:* "About to short these Korean fried chicken stocks. what is a korea"

- **Standard VADER**: Neutral (compound score: $+0.000$)
- **WSB-enhanced VADER**: Negative (compound score: -0.250)

Example 3 *Text:* "Grandma just died. I YOLO'd my inheritance into NVDA. NVDA to the moon! Diamond hands baby! "

- **Standard VADER**: Negative (compound score: -0.418)
- **WSB-enhanced VADER**: Positive (compound score: $+0.658$)

4.2 Ticker extraction

A stock ticker is a unique abbreviation (typically three or four letters) that is used to identify publicly traded shares of a stock Tickers can be extracted from the Reddit comments by the following methods:

1. A token preceded by `$` and matching `[A-Z]{1,5}`
2. An all-caps word of length 1–5 that is punctuation/space bounded and present in the NASDAQ symbol set.

4.3 Sentiment classification

Each Reddit comment is labeled using the VADER sentiment analyzer. VADER assigns a compound sentiment score in the range $[-1, 1]$, where -1 indicates extremely negative sentiment and $+1$ indicates extremely positive sentiment. The compound score is discretized into sentiment classes as follows:

- **Negative**: compound score < -0.05
- **Neutral**: $-0.05 \leq$ compound score ≤ 0.05
- **Positive**: compound score > 0.05

4.4 Signal Generation

For each date and ticker, we aggregate sentiment from all Reddit comments that mention the ticker on that date. Let c denote a comment and $S(c)$ its Reddit score (net upvotes). The net sentiment score N is defined as

$$N = \sum S(c) \mid c = \text{pos} \; - \; \sum S(c) \mid c = \text{neg}.$$

4.5 Backtest protocol

For each day d, we convert selected tickers into trades. Let p_d^{open} and p_d^{close} respectively denote the open and close prices of a ticker on day d. We enter positions at the next trading day's open (after the sentiment date) and exit after H trading days (the holding period). The return for a long position is

$$r^{\text{long}} = \frac{p_{d+H-1}^{\text{close}} - p_d^{\text{open}}}{p_d^{\text{open}}},$$

and for a short position,

$$r^{\text{short}} = \frac{p_d^{\text{open}} - p_{d+H-1}^{\text{close}}}{p_d^{\text{open}}}.$$

Daily portfolio returns are computed as

$$R = \frac{\sum_t w_t \, r_t}{\sum_t w_t},$$

where $w_t = s(t)$ is the sentiment signal for ticker t. The intuition is that people who upvote agree with the comment, creating stronger sentiment thus weight in the portfolio.

4.6 Benchmark comparison

We compute daily benchmark returns for SPY and MAG7 over the same date range as the trading strategies. We use the following evaluation metrics:

- Cumulative returns for trading off sentiment and benchmarks.
- Correlation matrices of daily returns with benchmarks for holding periods
- Directional accuracy (accuracy which the sentiment trading daily return direction matches SPY)

Directional prediction accuracy over N days for strategy X relative to SPY is defined as

$$\text{Acc} = \frac{1}{N} \sum_{d=1}^{N} \delta\big(\text{sign}(R_d^X),\ \text{sign}(R_d^{\text{SPY}})\big)\ ,$$

5 Results and Analysis

We evaluate sentiment-driven trading strategies derived from *r/wallstreetbets* (WSB) and *r/stocks* across holding periods of 1, 3, 7, and 14 trading days. Results are reported in terms of cumulative returns, equity curve dynamics, market alignment, and directional predictive accuracy.

5.1 Strategy Returns by Holding Period

Table 1 reports cumulative returns for each strategy and holding horizon. The r/stocks strategy exhibits negative returns at the 1 day holding period at -16.89% but shows increasing returns as the holding periods go up. This suggests that r/stocks sentiment captures long term fundamental information that requires time to be priced in. In contrast, the r/WSB community shows excellent returns with a 1 day hold window at 56.96% but collapses at the 3 day period falling to a mere -0.34%, indicating that r/WSB sentiment capture short term momentum that quickly reverts back to the mean. Interestingly however, as the holding period increases the r/WSB community returns improve, likely due to the community discussing the same retail-popular stocks as r/stocks but with different time framing. This shows that at longer horizons the underlying movement of the asset becomes increasingly more important than the initial sentiment signal.

Strategy	1 Day	3 Days	7 Days	14 Days
WSB	+56.96%	-0.34%	+18.87%	+45.35%
Stocks	-16.89%	+8.70%	+45.63%	+82.37%

Table 1: Cumulative returns by holding period.

5.2 Equity Curves by Holding Period

Figure 1 shows cumulative equity curves for both sentiment-based strategies across holding horizons. These results show distinct differences in the patterns of returns between the two communities. While both strategies are volatile compared to the SPY and MAG7 indexes, r/WSB is much more volatile than r/stocks, confirming the culture differences between the "YOLO" culture of r/WSB and more stable fundamentals backed approach of r/stocks. Furthermore, the convergence of both strategies as the holding period increases implies that the community-specific sentiment signals become less relevant over longer time horizons. Instead the difference lies in the timing of the trades. This creates the implication that the stocks discussed are correlated bets on a shared subset of retail-popular stocks that dominate the discussion among both communities.

5.3 Market Alignment Across Holding Periods

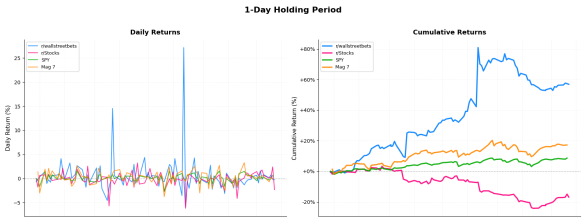
Table 2 reports the correlation between sentiment-based strategy returns and SPY across holding horizons. The extremely low correlations overall indicate that the sentiment from the subredits has little relationship with the broad market movements.

Strategy	1 Day	3 Days	7 Days	14 Days
WSB vs. SPY	0.132	0.061	0.016	0.046
Stocks vs. SPY	0.344	-0.002	0.183	0.174

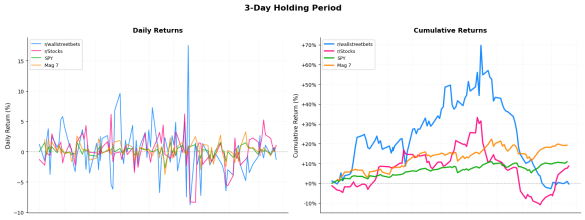
Table 2: Correlation of sentiment-based strategies with SPY across holding periods.

5.4 SPY Directional Prediction Accuracy

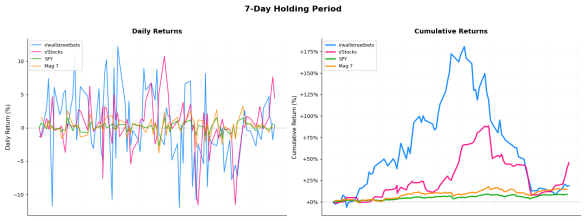
Table 3 reports the directional prediction accuracy of sentiment-based strategies with respect to SPY. For 1 day and 3 day holding periods, r/WSB performs around 50% accuracy, no better than flipping a coin. However this accuracy collapses to 37.1%, significantly worse than random which can be explained by the mean reversion of the characteristic short "YOLO" plays of r/WSB. At 14 days, accuracy recovers to 51.9%, suggesting that as the holding period significantly increases, the underlying asset's fundamental movement takes precedence over the timing of entry and exit trades, as described in sections 5.1 and 5.2. r/stocks, on the other hand sees as slight dip from 54.7% to 53.8% from 1 Day holding to 3 Day holding pe-



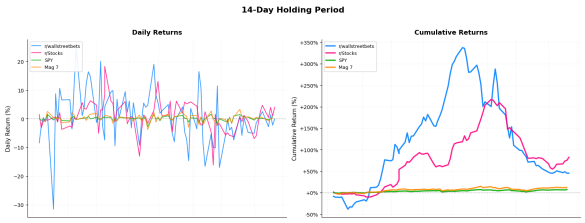
(a) 1-day holding period



(b) 3-day holding period



(c) 7-day holding period



(d) 14-day holding period

Figure 1: Cumulative equity curves for sentiment-based trading strategies across holding periods.

riods, but then significant gains in prediction accuracy with 58.4% at 7 days and 65.4% at 14 days. As a fundamentals-based community, this pattern makes sense: longer holding periods allow the strategy to better capture long-term market trends as fundamental analysis requires time to be reflected in prices.

Strategy	1 Day	3 Days	7 Days	14 Days
WSB	50.5%	49.5%	37.1%	51.9%
Stocks	54.7%	53.8%	58.4%	65.4%

Table 3: SPY directional prediction accuracy across holding periods.

6 Conclusion

6.1 Results

The study examined whether sentiment expressed in the subreddits or r/stocks and r/wallstreetbets can predict short term market movements. The analysis reveals the differences in culture of the communities, where r/wallstreetbets captures explosive short-term movements that quickly revert to the mean in a few days, seen in the good performances in the 1 day windows, and substantially worse performances in the 3 day holding period. r/stocks on the other hand identifies longer term trends that require time to materialize, with growing performance as the holding window increases. While a trading strategy off of sentiment analysis achieved substantial cumulative returns across holding periods, analysis shows that these come from exposure to retail popular stocks rather than reliable signals of market direction, where the cumulative return graphs converge as the holding period increases and the extremely low correlations of both strategies to SPY. These findings indicate that online sentiment may support profitable but risky trading strategies focused on a basket of volatile retail stocks, not as a reliable predictor for the broader market.

6.2 Related Work

Other papers have previously explored the relationship between sentiment in text, whether in news, social media, or online forums, and financial market movement. Early studies focused on news sentiment and found a small but statistically significant relationships with market returns and volatility (5). More recent work has look at how sentiment on social media platforms like Twitter and Reddit has weak predictive effects and/or issues with time horizons that change by asset and methodology (6). Several studies specifically focus on Reddit data, and they suggest that sentiment may simply capture investor attention rather than predicting broad market direction. (7). Our paper builds upon the current literature by comparing sentiment across specific and distinct Reddit communities, improving a rule and lexicon-based model through including community-specific language, and evaluating performance across multiple holding periods using both predictive and return-based metrics.

6.3 Further Improvements

The accuracy of the results rely on the assumption that the sentiment model is able to accuratley capture the sentiment of the comments. Further improvements for our project could include replacing VADER with a transformer-based sentiment

model, which would better capture contextual and intent-level sentiment in Reddit comments and posts. Because VADER is a rule-based model and is built on a dictionary with rule adjustments for negation, intensity, and punctuation, it does not model compositional meaning or semantics. Thus, a sentence like “might short INTC, looks like free money” may be classified as positive sentiment due to words like “free” and “money" in the sentence, even though the actual sentiment is negative (shorting a stock). Transformer-based models can understand the context of the full sentence, thus correctly associating the sentiment with the act of shorting the asset. Furthermore, if an comment contained a positive sentiment about one ticker and negative about another, in the current implementation, the full context would not be captured. Therefore using a transformer based model to extract tickers and relative sentiment would be more ideal.

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